

ENVS 193DS Final

Table of contents

Set Up	1
Problem 1. Research Writing	2
a. Transparent Statistical Methods	2
b. Suggestions for rewriting	2
Part 1.	2
Part 2.	2
c. Further Analysis	3
Problem 2. Data Analysis	3
a. Clean and Wrangle	3
b. Response Variable	4
c. Purpose of Study	4
Tree Height	4
Acacia Ant Species	4
d. Table of Models	4
e. Run the Models	5
f. Select the Best Model	5
g. Check the Diagnostics	6

[GitHub Repo](#)

Set Up

```
# loading in packages
library(tidyverse) # general use
library(janitor) # cleaning data
library(here) # organizing files
```

```
library(rstatix) # statistical relationships
library(MuMIn) # modelling
library(DHARMA) # residual diagnostic simulation
library(ggeffects) # model predictions
library(ggplot2) # visualizations

# read in data
nests <- read_csv(here("data", "nest_data_final.csv")) # read in and make object for nest data
dinner <- read_csv(here("data", "Dinner_Data.csv")) # read in and make object for personal data
```

Problem 1. Research Writing

a. Transparent Statistical Methods

In part 1, the researchers used a Generalized Linear Model (GLM) to reach their conclusion and compute that p-value, since the claim answers the central question of how distance to water edge predicts the probability of American Avocet microhabitat use. Additionally, since the variables are binary (use of microhabitat or no use) and continuous (distance from water edge) are non-parametric. Furthermore, since the response variable is binary we know the test was a logistic regression, a specific type of generalized linear regression. In part 2, they used a chi-square test because they are examining the relationship between two categorical variables, bird species and habitat type.

b. Suggestions for rewriting

Part 1.

We found that American Avocets do not use microhabitats in a random fashion, but rather with respect to distance to water edge. In this sense, distance to a water edge acts as a predictor for the likelihood American Avocet use a microhabitat.

Part 2.

When looking at the distribution of American Avocet, Forster's Tern, and Black-necked Stilt, we found that each species was not spread randomly among habitats. Rather, each species presence differed based on habitat. This conclusion takes into consideration managed pond, non-tidal marsh, salt pond, tidal marsh, and tidal mudflat as habitats.

c. Further Analysis

Another variable that could influence the probability of American Avocet microhabitat use is the predator abundance, which could be count data or categorical with pre determined levels from low presence to high presence of predators. This variable will influence American Avocet habitat use because this species might avoid areas with high concentrations of predators as they pose a risk for injury, mortality, and reduce reproductive success.

Problem 2. Data Analysis

a. Clean and Wrangle

```
# create new object, nest_clean
nests_clean <- nests |>
  # keep only height, bird species, and ant species
  select(height_cm, case_control, ant_species) |>
  # filter to include on C. mimosae, C. sjostedti, and C. nigriceps
  filter(ant_species %in% c("AB", "RRB", "BBR")) |>
  # order the species
  mutate(ant_species =
    as_factor(ant_species),
    ant_species = fct_relevel(ant_species,
                              "AB",
                              "RRB",
                              "BBR")) |>

# create new column for scientific species names
mutate(scientific_name = case_match(
  ant_species,
  "AB" ~ "Crematogaster sjostedti", #C. sjostedti
  "RRB" ~ "Crematogaster mimosae", # C. mimosae
  "BBR" ~ "Crematogaster nigriceps" # C. nigriceps
))

# display new structure of the data
str(nests_clean)
```

```
tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ height_cm      : num [1:270] 392 310 513 154 324 ...
 $ case_control   : num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ ant_species    : Factor w/ 3 levels "AB","RRB","BBR": 2 2 2 2 1 2 3 2 1 3 ...
```

```
$ scientific_name: chr [1:270] "Crematogaster mimosae" "Crematogaster mimosae" "Crematogaster mimosae"
```

```
# display 10 random rows from data  
slice_sample(nests_clean, n = 10)
```

```
# A tibble: 10 x 4
```

	height_cm <dbl>	case_control <dbl>	ant_species <fct>	scientific_name <chr>
1	146.	0	RRB	Crematogaster mimosae
2	159	0	RRB	Crematogaster mimosae
3	234	1	RRB	Crematogaster mimosae
4	128	0	BBR	Crematogaster nigriceps
5	194	0	RRB	Crematogaster mimosae
6	156	0	AB	Crematogaster sjostedti
7	279	0	RRB	Crematogaster mimosae
8	109	0	BBR	Crematogaster nigriceps
9	164.	0	BBR	Crematogaster nigriceps
10	204	0	RRB	Crematogaster mimosae

b. Response Variable

The 1s and 0s in the case_control column indicates whether or not a tree contained a nest. If a data point has a 1 this indicates the tree contains a nest, whereas a 0 indicates the tree did not contain a nest and was “available” for comparison.

c. Purpose of Study

Tree Height

Tree height is a numerical, continuous variable measured in cm. This variable can influence the probability of nest occurrence because as tree height increases

Acacia Ant Species

This is where I will describe the variable

d. Table of Models

Model number	Ant Species	Tree Height	Model description
0	X	X	All predictors (No Interaction term, +)
1	X	X	All predictors (Interaction term, *)

e. Run the Models

```
# model 0: all predictors, no interaction term
nest_model0 <- glm(case_control ~ ant_species + height_cm,
  # no interaction model variables
  data = nests_clean, # use clean dataset
  family = binomial) # set to binomial

# model 1: all predictors, interaction term
nest_model1 <- glm(case_control ~ ant_species * height_cm,
  # interaction model variables
  data = nests_clean, # use clean data set
  family = binomial) # set to binomial
```

f. Select the Best Model

```
# calculate AIC
AICc(
  nest_model0,
  nest_model1
) |>
  arrange(AICc)
```

```
      df    AICc
nest_model1  6 238.1236
nest_model0  4 258.8940
```

The best model for predicting the probability of bird nest presence as determined by Akaike's Information Criterion (AIC) is the model of interaction between tree height (cm) and ant species, demonstrated by the lower AIC value.

g. Check the Diagnostics

```
# check diagnostic with DHARMA
plot(
  simulateResiduals(nest_model1)
)
```

